

Spring Boot

Chapter - 1

→ Spring:

To build an enterprise or web application.

framework:

- ~~it~~ it provides an environment and tools to build applications.

- Spring contains classes & interfaces (JAR files).

* POJO → class file containing:

(Plain old Java Object)

→ private variables

→ getters & setters → annotations (@)

POJO

- creates & manages files using Spring configuration.
- All functions are written inside this file.

* Dependency Injection (DI):

- Objects are created and dependencies are injected automatically by Spring.

@Autowired → annotation used for creating objects without new.

* Rest API:

- Spring supports building Restful APIs.
- Used for transferring data between client and server.

* Model View Controller (MVC) → Main feature

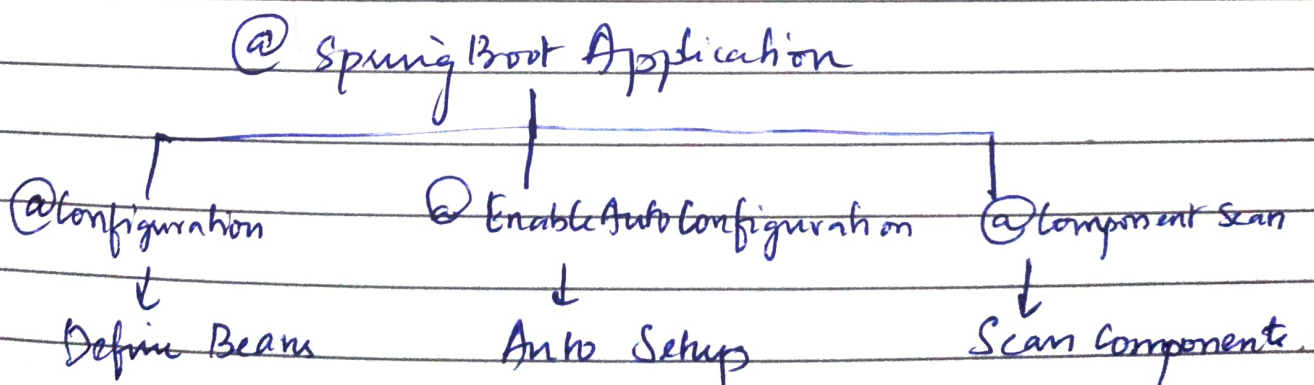
(1) Model → • defines the data structure
• Represents application data & business logic is handled here.

(2) View → • display data (UI / Frontend)

(3) Controller → • controls logic (uses request / response)
• connects model & view.

[User] → [controller] → [Model] → [View] → [User]

Maven, Tomcat, Apache → Spring initializes.
↓ ↓ ↓
Project Management Tool Deployment Tool (web server) Utilities library



1. @ Configuration → Defines Beans
2. @ EnableAutoConfiguration → Automatic Configuration
3. @ Component Scan → Scans components.

@ Component Scan

→ Spring automatically scans the packages for components like:

↳ @ Component

@ Service

@ Repository

@ Controller

@ RestController

@ Configuration

↳ This annotation tells the spring that the class contains the bean configuration.

↳ To define beans this is used.

@ EnableAutoConfiguration

↳ Automatically configures & sets up:

◦ web servers

◦ Database

◦ Security.